

Dispersion of immiscible liquids in a static mixer without inserted elements: A CFD-PBM study on droplet size reduction and distribution

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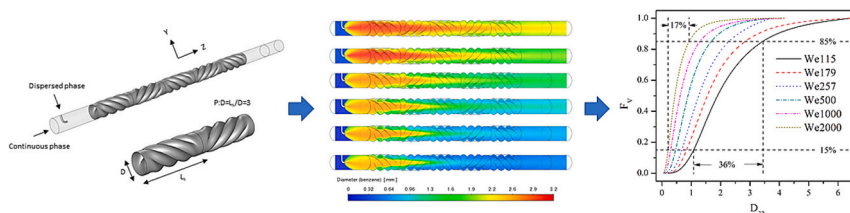
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HIGHLIGHTS

- A novel static swirl mixer was applied to immiscible liquids dispersion.
- The size reduction and droplet size distribution were evaluated by CFD-PBM method.
- Correlations characterizing swirl mixer's dispersion performance were studied.
- The energy-saving potential of the static swirl mixer was revealed.

GRAPHICAL ABSTRACT



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ABSTRACT

A CFD-PBM (Computational Fluid Dynamics-Population Balance Model) coupled model has been employed to investigate the turbulent dispersion of immiscible liquids through a static swirl mixer without inserted elements, aiming to mitigate the disadvantages of existing static mixers, e.g., high energy consumption and maintenance costs. The model was firstly validated in a Kenics static mixer with experimental data of a benzene-water system with a benzene volume fraction of 0.5 % and empirical correlations, and was then used to evaluate the performance of the proposed static swirl mixer in pressure losses, Sauter mean diameter, droplet size distribution, effect of number of swirl sections, and PD (pitch to diameter) ratios. The static swirl mixer demonstrated a clear effect in the dispersion of immiscible benzene into water producing fine droplets, with its dispersive mixing efficiency being more pronounced in flows with higher Weber numbers. Its droplet size reduction effect is relatively less effective than the Kenics mixer for the same Weber numbers, attributed to the absence of inserted elements, but at the expense of a much lower pressure loss than the Kenics mixer. In producing the same droplet sizes, the static swirl mixers cause a lower pressure loss than the Kenics mixer to a certain extent depending on different the Weber numbers and PD ratios, demonstrating an energy-saving potential. A decreasing PD ratio produces smaller droplet sizes but at higher pressure losses. A PD ratio between 1.5 and 3 may be optimal. Ten swirl sections would appear to be sufficient to produce the equilibrium droplet size distribution despite the different PD ratios.

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1. Introduction

Mixing of liquids is a key feature in most industrial processes covering a broad spectrum of applications in the chemical, pharmaceutical, cosmetic, and the food and beverages industries [1], either for in-phase mass transfer, chemical reactions, or simply blending components to make a product. The components may be miscible or immiscible fluids and can be dispersed in each other down to the molecular level to reduce the inhomogeneity of the system caused by a gradient of phase, viscosity, concentration, or temperature [2], with the issues being the mixing speed, the power required, the uniformity of mixing, and for immiscible liquids, the scale down to which dispersion can be achieved. The mixing processes are fundamental in determining a product's yield, quality, physicochemical properties, and reaction rate in a given process. In the meantime, mixing is an energy intensive process. This is especially true for the immiscible liquid-liquid dispersions as it demands high energy consumption for the breakup of dispersed phase to generate fine droplets, and for the sufficient blending of compositions to achieve a high level of composition uniformity [3].

Adopting in-line pipe mixers for the dispersion of one or more immiscible liquids into a continuous phase offers several advantages over batch processing and mechanically stirred vessels due to their small space requirement, reduced energy consumption, and sharp residence time distribution [4]. The in-line pipe mixer may be dynamic, with a rotating element inserted in a hollow tube, or may be static or motionless, with stationary inserted elements, typically helical blades or corrugated plates inserted in a hollow tube, to induce changes in the fluid streamlines [5]. In comparison to dynamics mixers, static mixers typically exhibit lower energy consumption and reduced maintenance cost as they do not have moving parts and therefore no external driving power and seals. They also require a smaller space, lower facility costs, and can be manufactured from most materials to meet various requirements [4]. Static mixers add extra pressure drop to the line in which they are located, using the energy of the stream itself to bring about mixing. By inserting elements in a pipe, the flow field is altered relative to that of an open pipe, splitting, mixing, and recombining two or more liquids together [6], yielding efficient mixing in a short conduit with no moving parts [5]. Nowadays static mixers are standard equipment items widely used in immiscible liquid-liquid dispersion to produce emulsified materials in the chemical, pharmaceutical, cosmetic, and food industries [7], though the conventional stirred vessels remain powerful tools especially for processing highly viscous products [8].

Static mixers consist of a series of inserted elements installed inside the pipes to promote a chaotic mixing behaviour. Based on their geometrical features and applications, the static mixers are classified into types of open design with helices, e.g. Kenics mixer, and EREstat mixer, open design with blades, e.g. HEV, LPD, corrugated plates, e.g. SMV, multi-layer design type, e.g. SMX, and other types. A detailed classification and common models can be referred to [1]. In this paper, we propose a static swirl mixer without inserted elements inside the pipe but by introducing turbulent swirling flows from the alternatively left and right twisted four-lobed pipe walls to facilitate the mixing process. This proposed design may be classified into the static mixer type 'open design with helices' as the geometrical features of the mixing elements are similar to that of a Kenics mixer, except that they are not inserted inside the pipes. Although the Kenics and the SMX mixers are the most common statics mixers in the industry, the SMX mixers are reported perform better than the Kenics mixers, especially in fluids with high viscosity, but also at the expense of higher pressure loss for the same length [9]. Given the absence of the inserted elements inside the proposed static swirl mixer, its applications are more likely to fall into the low to medium viscous products, similar to the Kenics mixers. With this in mind, the following literature review will focus on the investigation of Kenics mixers and their dispersion of immiscible liquids, to provide the knowledges for the validation and evaluation of the proposed static swirl mixer.

Kenics static mixers represent one of the classic types of static mixers for mixing fluids/granular materials, having the advantages of their unique structure and easy manufacturing [10–12]. It uses a series of mixing elements made from thin, flat strips, twisted through 180° to form helices having an aspect of 1.5 to 2.5 to systematically split, reorientate and recombine two liquids. In the most cited experimental work, Middleman [13] studied the dispersion of six organic liquids of different viscosity and interfacial tension with water through a Kenics static mixer with a pitch (L_0/D) of 1.5 and a thickness of 2.5 mm in 1 and 0.5 in. pipes. The Sauter mean droplet diameter D_{32} was observed to decrease with a larger Weber number, We , and lower viscosity and volume fraction. The effect of the number of elements ($n_e = 3, 6, 21$ respectively) was tested in a benzene-water system and it indicated that ten elements were sufficient to produce the equilibrium droplet size distribution. For dispersed phases with a viscosity less than $1 \times 10^{-3} \text{ kgm}^{-1} \text{ s}^{-1}$, the Sauter mean diameter D_{32} with Weber number was well correlated by

$$\frac{D_{32}}{D} = A We^{-3/5} Re^{0.1} \quad (1)$$

Where A is a dimensionless empirical constant, We is the Weber Number given by

$$We = \frac{\rho_c \bar{U}_c^2 D}{\sigma} \quad (2)$$

Where ρ_c is the continuous phase density, $\bar{U}_c$ is the continuous phase mean velocity, σ is the surface tension between the dispersed and continuous phases, D is the pipe diameter. The Weber number is a dimensionless number that describes the ratio of the disruptive forces that break the droplets and the surface tension that prevents the droplet breakup, during the emulsification process.

In addition to Middleman's work, Berkman and Calabrese [14] conducted experiments in a 19.1 mm diameter pipe with 24 Kenics elements, where silicone oil, paraffin oil, and p-Xylene with viscosities ranging from 0.63 to 204 mPa·s, and with volume fraction less than 0.1 %, were dispersed into water, and developed a semi-empirical theory that correlated the mean droplet size with the Weber number and calculated the Fanning friction factor, f , for Reynolds numbers ranging from 12,000 to 21,000 (Weber number covering 144 to 665). The correlation of the Sauter mean diameter D_{32} with Weber number is:

$$\frac{D_{32}}{D} = 0.49 We^{-0.6} \left(1 + 1.38 Vi \left(\frac{D_{32}}{D} \right)^{0.33} \right)^{0.6} \quad (3)$$

$$Vi = \frac{\mu_d \bar{U}_c}{\sigma} \sqrt{\frac{\rho_c}{\rho_d}}$$

Where Vi is a viscosity group or capillary number representing the ratio of viscous to surface forces acting to stabilize the droplet; μ_d is the dispersed phase viscosity; ρ_c is the continuous phase density, and ρ_d is the dispersed phase density.

The Fanning friction factor, f , of the Kenics static mixer by Berkman and Calabrese [14] is defined as:

$$f = \frac{\Delta P}{2\rho \bar{U}^2} \frac{D}{L} \quad (4)$$

Where ρ is the fluid density, and L is the Length of pipe corresponding to the pressure loss.

Additionally, Chen and Libby [15] conducted an experimental study on gas-liquid and liquid-liquid dispersions in a Kenics mixer and proposed a correlation to calculate the Sauter mean diameter D_{32} where

$$\frac{D_{32}}{D} = 1.14 We^{-0.75} \left(\frac{\mu_d}{\mu_c} \right)^{0.18} \quad (5)$$

For the numerical prediction of the droplet/bubble Sauter mean

diameter D_{32} , many research adopted a Computational Fluid Dynamics-Population Balance Model coupled approach to predict the droplet size distribution (DSD), which fit accurately the experimental data [10,16–19].

Despite the large body of literature on experimental and numerical investigations [10,11,20–23] on Kenics static mixers, the main focus has been on the optimization of the Kenics helices geometries, e.g., optimal angle of twist, the ratio of the element's length to its diameter, the offset angle between two consecutive elements, the effect of element thickness, etc. It does not fundamentally address the disadvantages of the mixing inserts compared to plain tubes: higher pressure drop, greater potential for fouling, the relative difficulty of cleaning, and greater costs of such devices [4].

In light of the above review, this paper proposes a static swirl mixer without inserted elements but by introducing turbulent swirling flows using swirl induction pipes to facilitate the mixing process. The swirl induction pipe imparts tangential and axial velocity components to the flow with a small pressure loss, stretching and redirecting the two fluids to disperse the immiscible liquids and reduce the inhomogeneity of the system. The swirl motion is geometrically induced to the flow by the spiral lobed cross-section of the swirl pipe which avoids the insertion of any mixing element which would otherwise be mounted inside the pipe, such as helical blades or corrugated plates, that would contribute to problems regarding the fouling and cleaning of the pipes. The swirl pipe has been proven to have a self-cleaning effect which will significantly reduce the potential of fouling and thus the cost of maintenance [24,25]. In this paper, we firstly developed a CFD-PBM model of Middleman's [13] Kenics mixer and validated it with Middleman's experimental data and Berkman and Calabrese's [14] and Chen and Libby's [15] correlations in terms of Sauter mean diameter D_{32} and pressure loss. We then used the identical validated CFD-PBM to predict and evaluate the performance of the static swirl mixer in terms of the pressure loss, Sauter mean diameter D_{32} , droplet size distribution, and the effects of the number of swirl sections and PD ratios on the dispersive mixing.

2. Numerical method and models

2.1. Computational domain

The computation domain of the Kenics mixer is as shown in Fig. 1a which is identical to Middleman's [13] experimental setup and Haddadi et al.'s [10] numerical study of the Kenics mixer. 10 helical blades with a length to diameter ratio of 1.5 and a thickness of 2.5 mm are used for the Kenics mixer. Each element is twisted through 180° , and alternate left- and right-hand twists are fixed in series down the axis of the pipe with a diameter of 25 mm. The plain pipe before and after the elements are 75 mm and 125 mm respectively. The static mixer is vertically upward. The continuous phase of water is entered upward through the 25 mm diameter circular pipe at the bottom and the dispersed phase of benzene is introduced from a 90° bend tube at the side with a diameter of 2.5 mm, facing the axis of the mixing elements. The water-benzene system in the Kenics mixer is selected as it has been extensively studied in the literature [10,13,14], providing sufficient experimental data and empirical correlations for validation purpose. The computational domain for the baseline static swirl mixer is shown in Fig. 1b which is the same as the Kenics mixer except that the mixing elements are replaced by 5 hollow swirl pipe sections with alternate clockwise and anticlockwise twists. Each swirl section has its 4-lobed cross-section twisted 360° in the length of 3 diameters resulting in a Pitch to Diameter (PD) ratio of 3, which has the same twist angle as the helical blades in the Kenics mixer. At the two ends of each swirl section, a transition section was designed to facilitate the gradual transition from a 4-lobed cross-section to a circular cross-section and vice versa. The transition sections reduce the pressure loss by avoiding sudden changes in pipe geometries. A detailed description of the swirl section can be found in [24,25]. The basis of the design of the static swirl mixer is the optimum balance between the dispersion

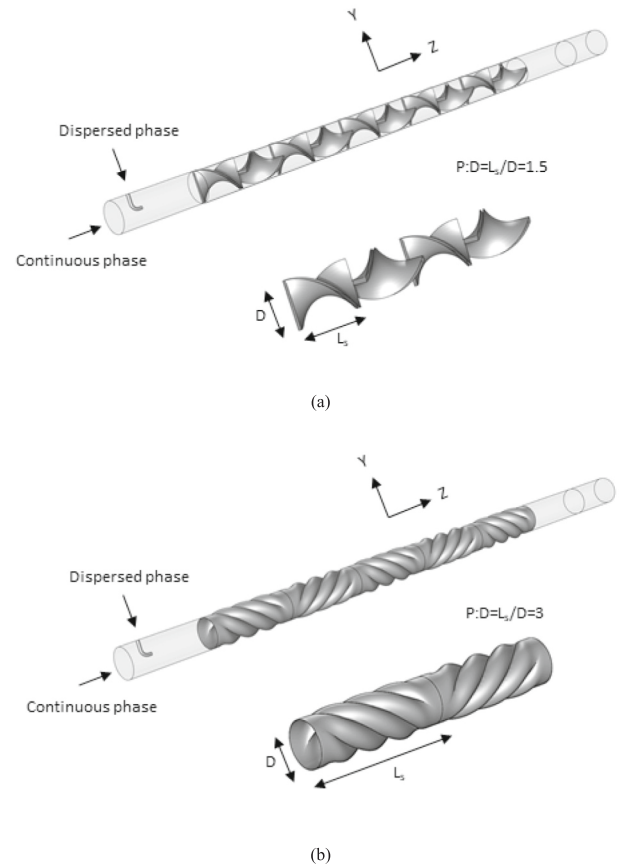


Fig. 1. Computational domain of the (a) Kenics mixer, and (b) static swirl mixer.

performance and the pressure loss occurred, that is to achieve the smallest droplet size with the narrowest distribution, with the expense of the minimum pressure loss. Therefore, besides the baseline design, we also examined the impacts of the number of swirl sections, and the different PD ratios on the dispersion performances and the corresponding pressure losses, to obtain insights into the optimum design.

2.2. Meshing

The ICEM CFD (ANSYS, USA) software was used for the discretization of the flow domain for both the Kenics and static swirl mixer using a hexahedral mesh with 5 layers of fine elements near the wall to capture the velocity gradient close to the walls. The mesh was then converted to polyhedral mesh in Fluent as shown in Fig. 2 which permits better

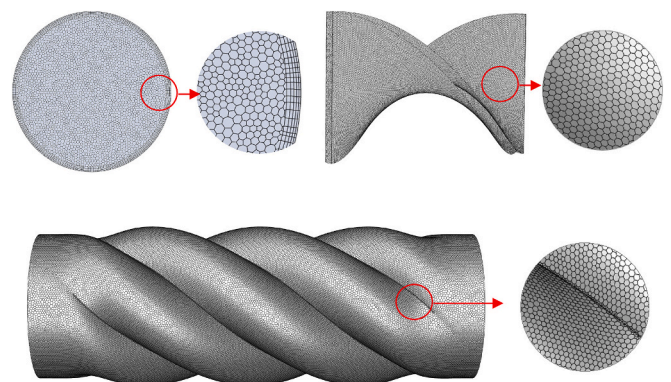


Fig. 2. Meshing of the Kenics mixer and static swirl mixer.

convergence of the solution and improves the computational efficiency [26]. A mesh dependence test was carried out with a maximum mesh size of 1.5 mm, 1 mm, and 0.5 mm respectively, corresponding to 40,1184, 1,279,658, and 9,768,937 grid elements respectively for the Kenics mixer. The water velocities at the outlet, the pressure drops over the Kenics mixer, and droplet sizes at the exit of the Kenics mixer were compared as shown in Table 1, and it suggested that the maximum mesh size of 0.5 mm ensures that the errors associated with the size of mesh is within 1 % and is therefore mesh independent.

2.3. Modelling turbulence

The turbulent flow within the mixers features a considerable degree of anisotropy in the stress and dissipation tensor leading to a highly anisotropic eddy viscosity [27]. Although many researchers adopted the $k - \omega$ turbulence model for modelling Kenics mixers [10,11], the most reliable Reynolds Stress Model (RSM) was often employed in studies involving swirl flows [25], and cyclones [28,29] for the accurate prediction of the central vortex and flow behaviour, as it accounts for the effects of streamline curvature, swirl, rotation, and rapid changes in the strain rate in a more rigorous manner than one-equation and two-equation models. It has greater potential to give accurate predictions for complex flows [30]. Due to the induction of strong swirl flows within the static swirl mixer, the RSM is employed in this study. The transport equation for the RSM is [31,32]:

$$\frac{\partial}{\partial t}(\rho u_i u_j) + \frac{\partial}{\partial x_k}(\rho u_k u_i u_j) = -\rho \left(\overline{u_i u_k} \frac{\partial u_j}{\partial x_k} + \overline{u_j u_k} \frac{\partial u_i}{\partial x_k} \right) - \frac{\partial}{\partial x_k} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial}{\partial x_k} (\overline{u_i u_j}) \right] + P \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \delta_{ij} \rho \epsilon \quad (6)$$

Where the terms on the left are local time derivatives and convections, the terms on the right are stress production, diffusion, pressure-strain and dissipation respectively. μ and μ_t are the dynamic and eddy viscosity, with δ_{ij} being the Kronecker delta function.

The pressure-strain term is modelled by

$$P \left(\frac{\partial y_i}{\partial x_j} + \frac{\partial y_j}{\partial x_i} \right) = -\rho \epsilon \left(c_{s1} a_{ij} - c_{s2} \left(a_{ik} a_{kj} + \frac{1}{3} a_{mn} a_{mn} \delta_{ij} \right) \right) - c_{r1} P a_{ij} + \rho k S_{ij} \left(c_{r2} - c_{r3} \sqrt{a_{ij} a_{ij}} \right) + c_{r4} \rho k \left(a_{ik} S_{jk} + S_{ik} a_{jk} - \frac{2}{3} a_{mn} S_{mn} \delta_{ij} \right) + c_{r5} \rho k (a_{ik} \Omega_{jk} + a_{jk} \Omega_{ik}) \quad (7)$$

where a_{ij} is the Reynolds-Stress anisotropy term, S_{ij} the strain rate and

Ω_{ij} the vorticity, with $a_{ij} = \frac{\overline{u_i u_j}}{k} - \frac{2}{3} \delta_{ij} S_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$, $\Omega_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right)$ and $P = (1/2) P_{kk}$, k is the turbulence kinetic energy, and ϵ the turbulence eddy dissipation rate computed from the additional transport equation

$$\frac{\partial \rho \epsilon}{\partial t} + \frac{\partial}{\partial x_k}(\rho u_k \epsilon) = \frac{\partial}{\partial x_k} \left[\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_k} \right] + \frac{\epsilon}{k} (c_{e1} P - c_{e2} \rho \epsilon) \quad (8)$$

The various constants used in this RSM are $\rho_k=0.68$, $\rho_\epsilon=1.3$, $c_{e1}=1.45$, $c_{e2}=1.83$, $c_{s1}=1.7$, $c_{s2}= -1.05$, $c_{r1}=1.7$, $c_{r2}=0.9$, $c_{r3}=0.8$, $c_{r4}=0.625$, $c_{r5}=0.2$.

At the near wall region, the enhanced wall treatment approach is used which considers other effects such as pressure gradients.

2.4. Eulerian multiphase model

In the Eulerian-Eulerian approach, the multiphase is treated as an interpenetrating continua medium which means that both of the phases are part of the computational domain and interpenetrate each other while they are moving [33]. As the volume of each phase cannot be occupied by the other phase, the volume fraction is introduced with the sum of the volume fraction for each phase equalling one. The continuity equation for phase q is [30].

$$\frac{\partial}{\partial t}(\alpha_q \rho_q) + \nabla \cdot (\alpha_q \rho_q \vec{v}_q) = \sum_{p=1}^n (\dot{m}_{pq} - \dot{m}_{qp}) + S_q \quad (9)$$

where ρ_q is the density of phase q , t is the time, $\vec{v}_q$ is the flow velocity of phase q , α_q is the volume fraction of phase q , $\dot{m}_{pq}$ characterizes the mass transfer from the p^{th} to q^{th} phase and $\dot{m}_{qp}$ characterizes the mass transfer from phase q to phase p , and S_q is the source term.

The momentum balance for phase q yields

$$\begin{aligned} \frac{\partial}{\partial t}(\alpha_q \rho_q v_i) + \nabla \cdot (\alpha_q \rho_q \vec{v}_q \vec{v}_q) = & -\alpha_q \nabla p + \nabla \cdot \bar{\tau}_q + \alpha_q \rho_q \vec{g} \\ & + \sum_{p=1}^n (\vec{R}_{pq} + \dot{m}_{pq} \vec{v}_{pq} - \dot{m}_{qp} \vec{v}_{qp}) \\ & + (\vec{F}_q + \vec{F}_{lift,q} + \vec{F}_{wl,q} + \vec{F}_{vm,q} + \vec{F}_{td,q}) \end{aligned} \quad (10)$$

Where $\bar{\tau}_q$ is the q^{th} phase stress-strain tensor:

$$\bar{\tau}_q = \alpha_q \mu_q \left(\nabla \vec{v}_q + \nabla \vec{v}_q^T \right) + \alpha_q \left(\lambda_q - \frac{2}{3} \mu_q \right) \nabla \cdot \vec{v}_q \vec{I} \quad (11)$$

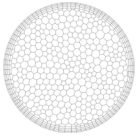
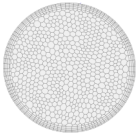
Where μ_q and λ_q are the shear and bulk viscosity of phase q , $\vec{F}_q$ is the external body force, $\vec{F}_{lift,q}$ is the lift force, $\vec{F}_{wl,q}$ is the wall lubrication force, $\vec{F}_{vm,q}$ is the virtual mass force, and $\vec{F}_{td,q}$ is the turbulent dispersion force, $\vec{R}_{pq}$ is an interaction force between phases, p is the pressure shared by all phases, and $\vec{v}_{pq}$ is the interphase velocity.

ANSYS Fluent uses the following interaction term for the interaction force $\vec{R}_{pq}$:

$$\sum_{p=1}^n \vec{R}_{pq} = \sum_{p=1}^n K_{pq} (\vec{v}_p - \vec{v}_q) \quad (12)$$

Where $K_{pq} (= K_{qp})$ is the interphase momentum exchange coefficient, and $\vec{v}_p$ and $\vec{v}_q$ are the phase velocities. Here, only the drag coefficient is considered and the Schiller-Naumann drag coefficient [34] is employed for the water and benzene droplets interaction.

Table 1
A summary of the mesh independence test results.

	Max size 1.5 mm	Max size 1 mm	Max size 0.5 mm
Model parameters: Kenics mixer, Weber number 500, Identical models and boundary conditions			
No. of elements	40,1184	1,279,658	9,768,937
Velocity at outlet, m/s	1.0142 (+1.91 %)	1.0036 (+0.84 %)	0.9952
Pressure drop of mixer, Pa	18,370.34 (+2.27 %)	18,127.71 (+0.92 %)	17,962.45
Mean D ₃₂ at mixer outlet, mm	0.9172 (+2.15 %)	0.9065 (+0.96 %)	0.8979

2.5. Population balance model

The immiscible liquid-liquid mixing process involves the dispersion of a secondary phase with a size distribution subject to the phenomena of aggregation and breakage. Here the Population Balance Model (PBM) was used incorporating the Eulerian model to predict the DSD within the static mixers. The PBM is one of the concepts in collective phenomenology. The concept is used when components or particles are experiencing birth and death processes where current particles are destroyed, and new particles are produced. The method solves a population balance equation for a number density function, which is the number of particles per unit volume within a range of values, normalized by the range. The evolution of the number density function is described in terms of effects of density and phase changes, coalescence and breakup etc. The number density function is treated as a continuous function corresponding to the limit of interval ranges to zero. The method of classes, together with suitable models for coalescence and breakup, are used to solve the number density function. The method of classes split the number density function into a number of intervals, or size groups, with each group assigned a single representative particle volume. The number density function can then be integrated over any subinterval to yield the number concentration of particles in the corresponding size group. In ANSYS Fluent, the population balance model is provided as an add-on module to solve flows involving a secondary phase with a size distribution that can evolve in conjunction with transport and chemical reaction in a multiphase system.

In the PBM in ANSYS Fluent, a number density function $n(\vec{x}, \phi, t)$ is used to account for the particle population, where different particles in the population can be distinguished and their behaviour can be described with the aid of particle properties. Here $\vec{x}$ is the external coordinates of a particle's state vector that denotes the spatial position of the particle, and ϕ is the internal coordinates which would include particle size, composition, and temperature. The average number of particles in an infinitesimal volume $dV_{\vec{x}}dV_{\phi}$ is $n(\vec{x}, \phi, t) dV_{\vec{x}}dV_{\phi}$ then the total number of particles in the entire system is [35].

$$\int_{\Omega_{\phi}} \int_{\Omega_{\vec{x}}} n dV_{\vec{x}} dV_{\phi} \quad (13)$$

The local average number density in physical space is

$$N(\vec{x}, t) = \int_{\Omega_{\phi}} n dV_{\phi} \quad (14)$$

The total volume fraction of all particles is

$$\alpha(\vec{x}, t) = \int_{\Omega_{\phi}} n V(\phi) dV_{\phi} \quad (15)$$

Where $\phi \in \Omega_{\phi}$, $\vec{x} \in \Omega_{\vec{x}}$, $V(\phi)$ is the volume of a particle in state ϕ .

Assuming that ϕ is the particle volume, the transport equation number density function $n(\vec{x}, \phi, t)$ is given by

$$\begin{aligned} \frac{\partial}{\partial t} [n(V, t)] + \nabla \cdot [\vec{u} n(V, t)] + \nabla_V \cdot [G_V n(V, t)] \\ = \frac{1}{2} \int_0^V a(V - V') n(V - V', t) n(V', t) dV' \\ - c \int_0^{\infty} a(V, V') n(V, t) n(V', t) dV' \\ + \int_{\Omega_V} p g(V) \beta(V|V') n(V', t) dV - g(V) n(V, t) \end{aligned} \quad (16)$$

Where G_V is the growth rate based on particle volume and $\nabla_V \cdot [G_V n(V, t)]$ the growth term. $a(V, V')$ is the aggregation kernel. $g(V)\beta(V|V')$ is the breakage kernel, where $g(V)$ is the breakage frequency, $\beta(V|V')$ is the probability density function of particles breaking from volume V to a particle volume of V , p is the number of child particles produced per parent. The terms on the right of the

equation are therefore birth due to aggregation, death due to aggregation, birth due to breakage, and death due to breakage respectively.

In this study, the Luo and Svendsen model [36] with a discrete solution method for PBM [35] was employed to calculate the breakage and coalescence kernels. In the discrete method, the particle population was discretized into a finite number of size intervals and the particle size distribution was calculated directly. This method is particularly useful when the range of the particle sizes is known a priori and does not span more than two or three orders of magnitude. With this method, the population can be discretised with a relatively small number of size intervals, thus is less computationally expensive. The Sauter mean diameter D_{32} of the benzene droplets was evaluated with

$$D_{32} = \frac{\sum_{i=1}^N n_i d_i^3}{\sum_{i=1}^N n_i d_i^2} \quad (17)$$

2.6. Boundary conditions

The continuous phase is water with a density of 998.2 Kg/m³ and a viscosity of 0.00089 Kg/(m·s), whereas the dispersed phase is benzene with a density of 870 Kg/m³ and a viscosity of 0.0006 Kg/(m·s). The surface tension between water and benzene is 0.04 N·m⁻¹. The dispersed phase volume fraction is $\alpha_d = 0.005$, which is a very dilute multiphase system. The above materials and properties are identical to Middleman's pioneering experimental work [13] for validation purpose. This study covers the Weber numbers range of 12 to 2000, which corresponds to the continuous phase Reynold numbers (Re) ranging from 3800 to 50,000.

The water entered the Kenics and static swirl mixers through the 25 mm diameter inlet with an "velocity inlet" boundary condition where its velocity was calculated by Eq. (2), and its turbulence was specified in terms of turbulent intensity $I = 0.16 \times (Re)^{-1/8}$ and hydraulic diameter at the inlet. A user-defined function (UDF) was implemented at the water inlet to provide a fully developed turbulent pipe flow where [37].

$$U_w = 1.224 \bar{U}_w \left(1 - \frac{r}{R}\right)^{0.14} \quad (18)$$

The benzene was injected into the mixers through the 2 mm diameter inlet with a uniform velocity calculated based on the volume fraction of $\alpha_d = 0.005$ and its area. The initial droplet size of benzene at its inlet was about 2 mm. In the PBM model, the discrete method with seven bins was used and the droplet size classes used in the simulation are listed in Table 2. The reason for choosing seven bins is that previous studies used seven to ten bins for simulating bubble or droplet size distribution and obtained reasonable results [10,38,39]. Considering the large number of cases need to be simulated in this study, and the large amount of additional computing time for more bins, the selection of seven bins should provide a reasonable balance between accuracy and efficiency. In Table 2, the range of the Sauter mean droplet diameter at different Weber numbers were estimated from Eq. (3) [14] and Eq. (5) [15] to allow droplet aggregation and breakage during the mixing. Basically, for a given Weber number, a mean droplet diameter can be estimated from the eqs. A ratio exponent, a value typically between 2 and 3, is be set between each neighbouring droplet sizes to meet the requirements that, the maximum and minimum droplet size does not span more than two or three orders of magnitude, and that the estimated mean droplet size should fall in the intermediate region of the bin sizes.

Table 2
Droplet size classes of the benzene used in the simulation.

Weber number	Droplet diameter ($\times 10^{-3}$ m)						
	Bin6	Bin5	Bin4	Bin3	Bin2	Bin1	Bin0
12–50	0.75	1.09	1.57	2.27	3.29	4.76	6.89
50–200	0.2	0.36	0.63	1.13	2	3.59	6.4
200–500	0.1	0.18	0.33	0.61	1.1	2	3.68
>500	0.05	0.1	0.22	0.46	0.96	2	4.22

A pressure outlet boundary condition with zero uniform static pressure was imposed at the outlet of the computational model. The turbulence was specified in terms of intensity and hydraulic diameter at the outlet in the same way as the inlets. The pipe walls were specified as being stationary and no-slip walls. The convergence criterion used for all cases was that the scaled residuals of x , y , z velocities, k , ϵ , benzene fractions and Reynolds stresses, etc. have decreased by five orders of magnitude.

3. Results and discussion

3.1. Validation of the CFD-PBM model

3.1.1. Friction factor

Pressure drop is a relevant indicator of the efficacy of static mixers as it indicates the pumping energy required for the mixing and thus the incurred costs [40]. Moreover, the pressure drop can be used as an adequate criterion to evaluate the correctness of the numerical simulations such as CFD [41]. To validate the CFD-PBM model, the Fanning friction factor through the Kenics static mixer is calculated with Eq. (4) and compared with Berkman and Calabrese's [14] measured data. As shown in Fig. 3, the calculated Fanning friction factors are in good agreement with the experimental data with a mean relative error (MRE) of 4.9 %. With increasing Reynolds number, the Fanning friction factor levels off, resulting in a roughly constant value of $f = 0.401$, showing good consistency with Berkman and Calabrese's [14] measured value of 0.416.

3.1.2. Droplet Sauter mean diameter

The droplet Sauter mean diameter for the Kenics mixer over Weber number 12, 17, 40, 80, 179, 257, and 500 are shown in Fig. 4 where the Sauter mean diameters at the plane 3D downstream of the Kenics mixer are 3.01 mm, 2.42 mm, and 1.54 mm for Weber numbers 12, 17, and 40, respectively. As the initial droplet diameter of benzene at its inlet is 2 mm, Fig. 5 indicates that, at a Weber number between 17 and 40, the droplet breakage starts to dominate. The predicted Sauter mean diameter D_{32} is compared with Middleman's [13] experimental data in Fig. 5a in the form of logarithmic D_{32}/D against logarithmic Weber number, the same form Middleman used, where D is the pipe diameter of 25 mm. The CFD-PBM model overestimates the droplet diameter by a maximum of 7.1 % and a mean relative error of 4.4 % in the Weber

number range of 12 to 40 in comparison to the experimental data, displaying good agreement.

A broader comparison of the predicted D_{32} with the Berkman and Calabrese [14] empirical correlation for estimating the D_{32} in the Kenics static mixers is shown in Fig. 5b, which covers Weber numbers from 12 to 2000. The mean relative error of the predicted D_{32} to the Berkman and Calabrese correlation is 9.1 %. In general, the CFD-PBM prediction is considered in good agreement with Berkman and Calabrese's correlation, and the inconsistency might due to the different configurations of the study, e.g., the different thickness of the element between the simulation model and the one operated by Berkman and Calabrese (2.5 mm vs 3.3 mm), the different number of elements (10 vs 24), the difference in pipe diameters (25 mm vs 19.1 mm), and the different Weber number range (12–2000 vs 144–665). In spite of this, there is good agreement in terms of the friction factor and the Sauter mean diameter of the benzene droplet, with the experimental data and empirical correlations in the literature. On this basis, the CFD-PBM model was deemed to be valid for applying to investigate the performance of the static swirl mixer.

3.2. The dispersion performance of the static swirl mixer

3.2.1. Droplet size reduction

The dispersive mixing of the static mixers induces droplet breakage due to the influence of turbulent eddies, producing fine droplets with narrow distributions. This section outlines the static swirl mixer's droplet size reduction effects in terms of the Sauter mean diameter D_{32} . Contour plot of the Sauter mean diameter of droplets for the static swirl mixer over a range of Weber numbers are shown in Fig. 6. Here, at a Weber number of 40, the disruptive force that the static swirl mixer exerted on the benzene droplets is not enough to overcome the surface tension force that prevents the droplet from breaking, resulting in droplet coalescence along the length of the swirl sections. By increasing the Weber number to 80, a balance in the forces between the droplet breakage and the coalescence rates is reached where the droplet size basically remains unchanged. It should be noted that, at We40 and We80, the predicted final droplet diameter will be 12 % and 32 % larger, respectively, when the 5 swirl sections are replaced by plain, circular pipes. This may suggest that the static swirl mixer plays a positive role in droplet size reduction even at a Weber number as small 40.

At We115 onward, the final droplet size decreases with increasing Weber number and Reynolds number, as larger droplet breakage rates are promoted by a stronger turbulent dissipation rate induced by the alternate left- and right-rotation of the swirl sections. It can be observed that the droplet size reduction effects are more prominent in the near-wall, lobed region where the droplet diameter is smaller, especially in the first few swirl sections. This may be attributed to a secondary wall jet-type tangential velocity [24,25] produced by the swirl sections where angular momentum is induced, concentrated in the near wall region, producing increased shear between droplets and eddies. The droplet diameter in the core region of the first few swirl sections, as depicted in red, is relatively large as the pipe is essentially "open" with no elements inserted inside. However, along the center line of the pipe the larger droplet diameters decrease in size with increasing distance down through the swirl sections. Besides the contribution of the swirl lobes, the droplet size reduction effect in this core region is enhanced by the turbulence caused by the sudden change of swirl rotation direction, from clockwise to anticlockwise and vice versa, where the fluid is further redistributed in the radial and tangential directions transverse to the main flow, being recombined, re-orientated and split. The droplet diameters reduce and become more uniform in cross-sections in a sequential fashion with increasing swirl sections and Weber numbers, displaying a prominent dispersive effect.

To better understand the droplet coalescence and breakage within the static swirl mixer at different Weber numbers, the radial variation of the Sauter mean diameters at the planes of the entrance of the first swirl

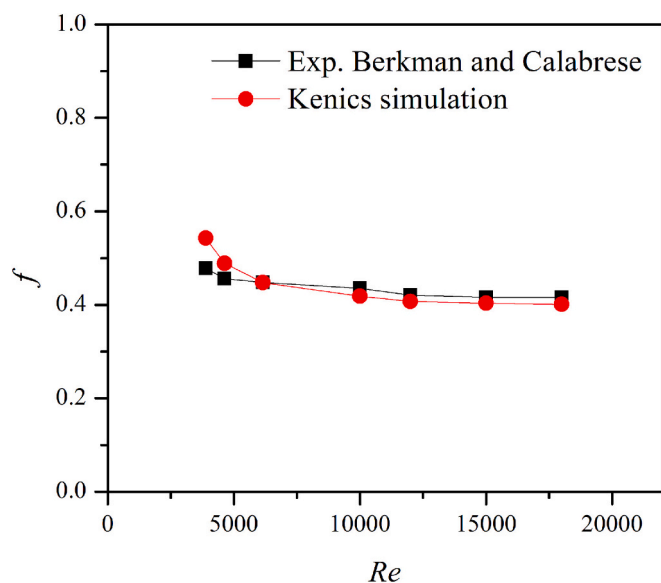


Fig. 3. Comparison of Fanning friction factor inside the Kenics static mixer with experimental data.

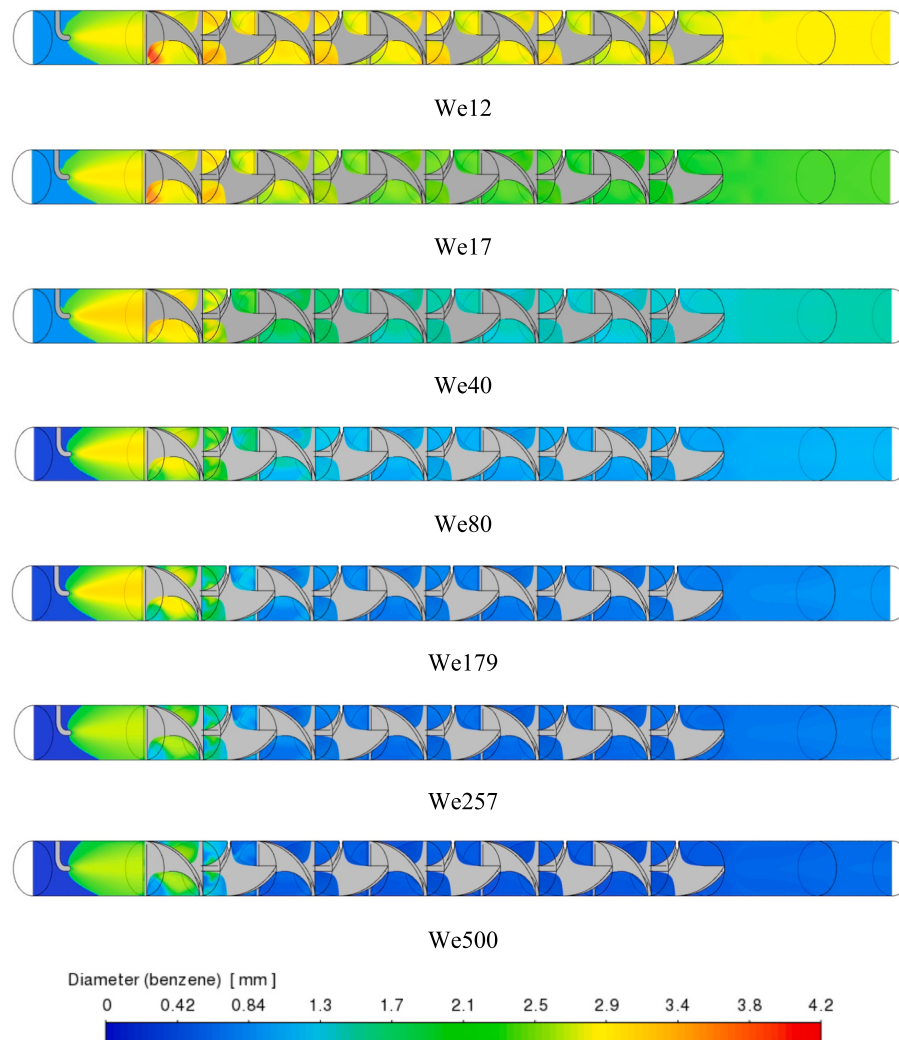


Fig. 4. Contour plot of benzene droplet Sauter mean diameter along Kenics static mixer at $We = 12, 17, 40, 80, 115, 179, 257$ and 500 respectively from top to bottom.

section, at each swirl section exit, and at 3D downstream of the last swirl section is shown in Fig. 7. The x axis in Fig. 7 is the pipe diameter in the form of r/R , where 0 indicates the pipe centre, and -1.0 and 1.0 are the pipe wall. At a small Weber number, e.g., $We17$, the droplet coalescence dominates and the droplet diameter increases with increasing swirl sections. From $We115$ onward, the droplet size reduces with increasing swirl sections and Weber numbers, as can be observed from the growing gap between the entrance of the first swirl section and 3D downstream of the last swirl section. It seems that stronger droplet coalescence and breakage occurs in the lobed wall region, displaying a sharper droplet size gradient than that in the pipe centre. This is more prominent with higher Weber numbers, especially in the first three swirl sections, where a stronger size reduction effect is observed in the lobed wall region. The radial distribution of the Sauter mean diameter is more evenly displayed in the plane 3D downstream of the static swirl mixer.

A comparison of the final droplet Sauter mean diameter at 3D downstream of the Kenics mixer, the static swirl mixer, and a circular plain pipe (replacing the mixer, and of the same length) is presented in Fig. 8. Following Middleman's [11] method, it is convenient to plot D_{32}/D as a function of the Weber number in the logarithmic form, where D is the pipe diameter of 25 mm. The dashed line represents the initial benzene droplet size at its injection point. It depicts that droplet breakage starts to dominate at a Weber number of approximately 25, 115, and 1000 for the Kenics mixer, the static swirl mixer and the plain

pipe respectively. To trigger size reduction effects, the static swirl mixer reduces the Weber number from 1000 to 115 in comparison to the plain pipe flow without any mixer. More importantly, thorough mixing in turbulent open pipe flow requires pipe lengths in excess of 100 pipe diameters, a distance often not available in chemical process industries [5]. The static swirl mixer, which is essentially an open pipe, makes it possible to mix with a much lower Weber number and shorter length of 15D. From Fig. 8, although the size reduction effect starts at $We115$ for the static swirl mixer, its positive impact on droplet breakage appears at a Weber number as small as 40 in comparison to the condition without any mixer. The final droplet size in the plain pipe flow is 12 % (at $We40$) to 279 % (at $We2,000$) larger than that downstream of a static swirl mixer, with the difference more prominent in flows with larger Weber numbers. At $We2,000$, the mean benzene droplet diameter downstream of the static swirl mixer is 0.43 mm, which is 21.5 % of its original size at its injection, demonstrating a clear dispersive mixing effect.

Despite the clear size reduction effects of the static swirl mixer, it is less effective than the Kenics mixer, accounting for 46 % (at $We115$) to 81 % (at $We2,000$) of size reduction effects at the same Weber number. This is expected as there are no inserted mixing elements inside the static swirl mixer, which primarily relies on turbulent swirling flow, induced through the spiral geometry of the swirl sections, for the mixing.

It is interesting to observe that, at $We40$ onward, the gap between the Kenics mixer and the static swirl mixer is almost constant in a log-log

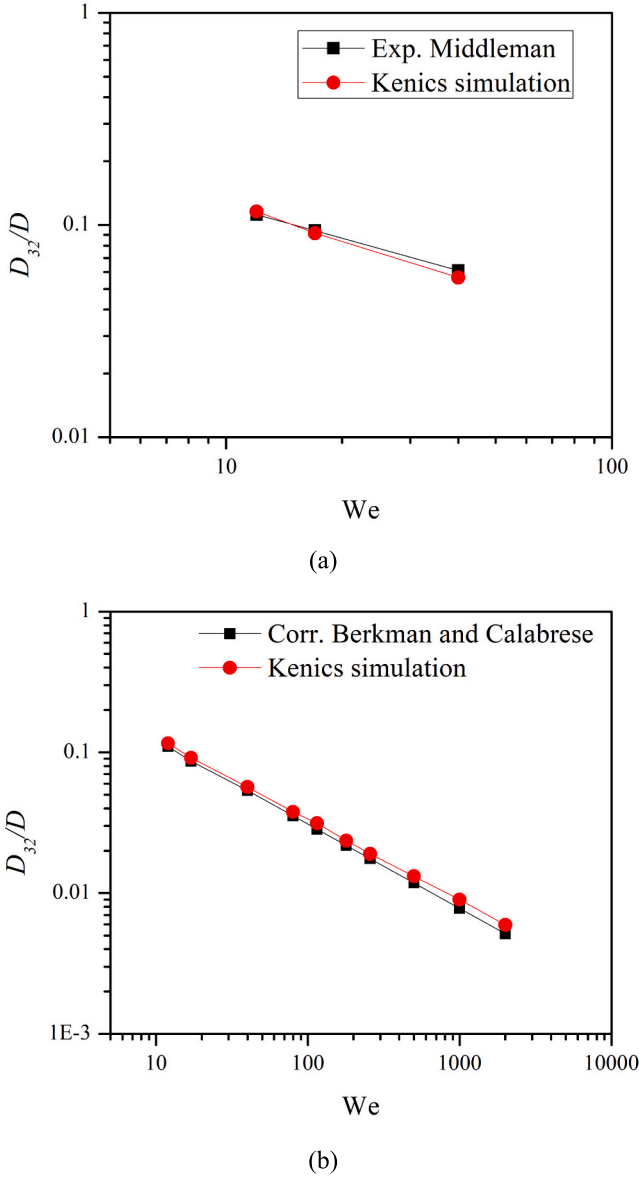


Fig. 5. Comparison of predicted D_{32} in Kenics static mixer with (a) experimental data and (b) empirical correlations, for benzene dispersion in water.

graph representation, as can be seen by the two parallel lines in Fig. 8. This suggests that the relationship between the Sauter mean diameter, D_{32} , and the Weber number in the static swirl mixer may fit a similar form of empirical correlation as the Kenics mixer. Data fitting indicates that this is indeed the case, where the droplet sizes and Weber numbers fit a power law correlation well for the static swirl mixer, with an absolute mean relative deviation of 4.17 %:

$$\frac{D_{32}}{D} = 0.885We^{-0.514} \quad (19)$$

Compared to the Berkman and Calabrese [14], and the Chen and Libby [15] correlations in Eqs. (3) and (5) respectively, the factors relating to liquid viscosity are missing in this power law correlation as the data is only from the water-benzene system with fixed physical properties. In the inviscid limit ($Vi \rightarrow 0$), the Berkman and Calabrese correlation for the Kenics mixer reduces to [14]:

$$\frac{D_{32}}{D} = 0.49We^{-0.6} \quad (20)$$

The correlation for the static swirl mixer is therefore similar to that of the Kenics mixer though less effective in droplet size reduction for the same Weber numbers.

3.2.2. Droplet size distribution

The Sauter mean diameter is a characteristic measure of the averaged diameter of the droplets that the static mixer produced, it does not indicate the range of the diameters produced and their proportion. In the population balance model, the droplet sizes were classified into 7 discrete groups as shown in Table 2, and the number density of each size were calculated using the breakage and coalescence kernels. As the size interval for classification is small compared to the range, it is reasonable to present volume fraction f_v as a continuous distribution. The volume fraction of each size is defined as [13].

$$f_v(D_i)dD_i = n_i D_i^3 / \sum_{i=1}^M n_i D_i^3 \quad (21)$$

The Cumulative Distribution Function (CDF) $F_v(D_k)$, is defined as below, which is the total volume fraction associated with the droplets of diameter less than some diameter, D .

$$F_v(D_k) = \sum_{i=1}^{i=k} f_v(D_i) \quad (22)$$

The volume fractions of the discrete droplet sizes were calculated at the planes of the exit of each of the 5 swirl sections using Eq. (21) for various Weber numbers and are shown in histograms in Fig. 9. The volume fractions of larger sizes decrease with an increasing number of swirl sections for all Weber numbers, and consequently the volume fractions of the smaller sizes increase. The Gaussian fit was attempted which suggests that the volumetric DSD exhibits a normal distribution after the third swirl section, with the peak of the curve roughly located at the mean of the Sauter mean diameters of the droplets at that plane. The distribution curve narrows and shifts towards the smaller sizes with increasing Weber numbers, characterizing a smaller mean droplet size and smaller standard deviation.

Fig. 10 plots the cumulative distribution function of the final droplet size at the plane 3D downstream of the static swirl mixer using Eq. (22). It shows that the breadth of the final DSD narrows with increasing Weber number, and thus energy dissipation rate. It can be estimated from the plot that, for a smaller Weber number of 115, 70 % of the volume of the dispersion is associated with droplets whose diameter lie within roughly 36 % of its droplet size range. While, with increasing Weber number to 2000, 70 % of the volume of the droplets lie within, roughly, 17 % of its droplet size range. This DSD is comparable to that of a Kenics static mixer reported by Middleman [13] and can be deemed relatively narrow.

To correlate droplet size distribution, Wang and Berkman [42] and Berkman and Calabrese [14] considered a relative resistance to breakage defined as the ratio of cohesive to disruptive forces acting upon a droplet. For a given droplet diameter, the relative resistance to breakage increases with dispersed phase viscosity μ_d and surface tension σ and decreases with Reynolds number or the mean energy dissipation rate. Therefore, the probability density function describing the distribution should depend on this relative resistance to breakage. Employing this argument, Wang and Berkman [42] correlated the equilibrium droplet size distribution, produced in stirred tanks, by normalization with D_{32} . Since their droplet size distributions were about normally distributed in volume, the normalized frequency was approximated by

$$F_v(X) = 0.5 \left[1 + \operatorname{erf} \left(\frac{X - \bar{X}}{\sqrt{2} \sigma_v} \right) \right] \quad (23)$$

Where $X = D/D_{32}$, and $\bar{X}$ and σ_v are the mean and the standard deviation respectively.

This equation was later used by Berkman and Calabrese [14] to

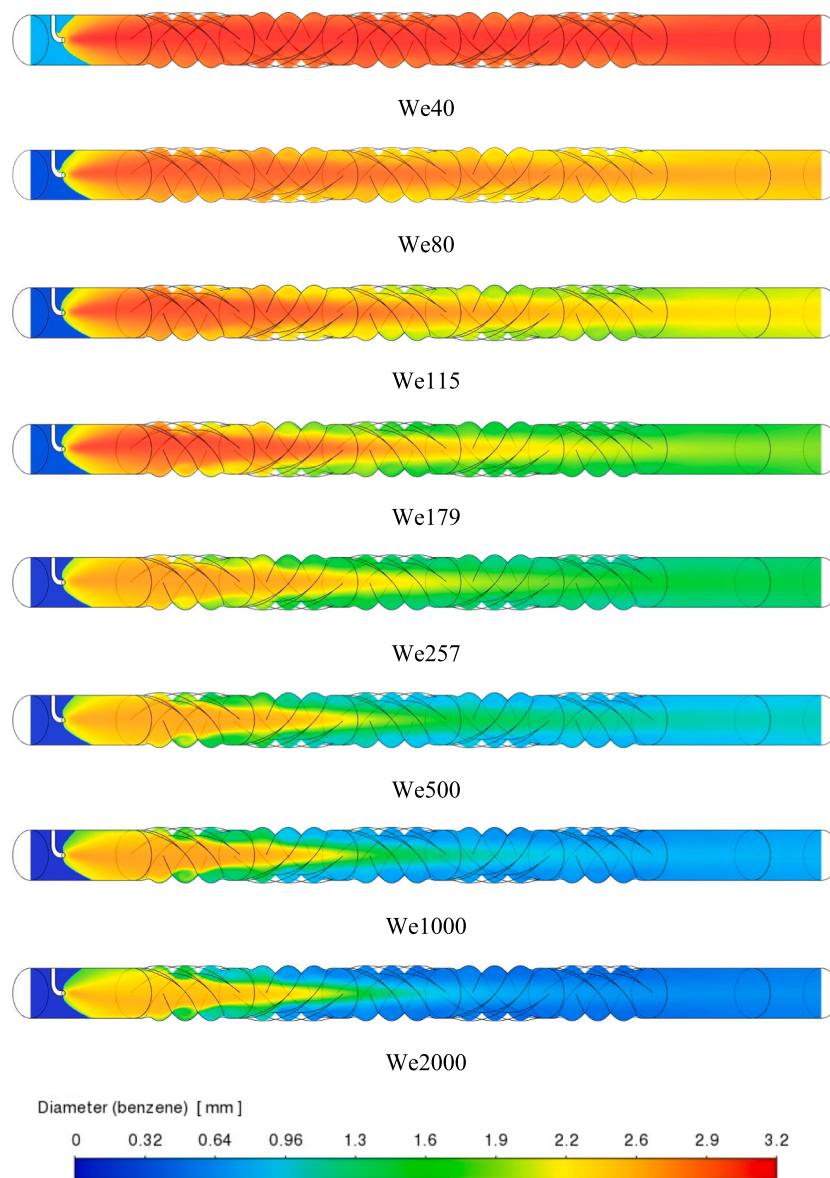


Fig. 6. Contour plot of benzene droplet diameter along static swirl mixer at $We = 40, 80, 115, 179, 257, 500, 1000$ and 2000 respectively from top to bottom.

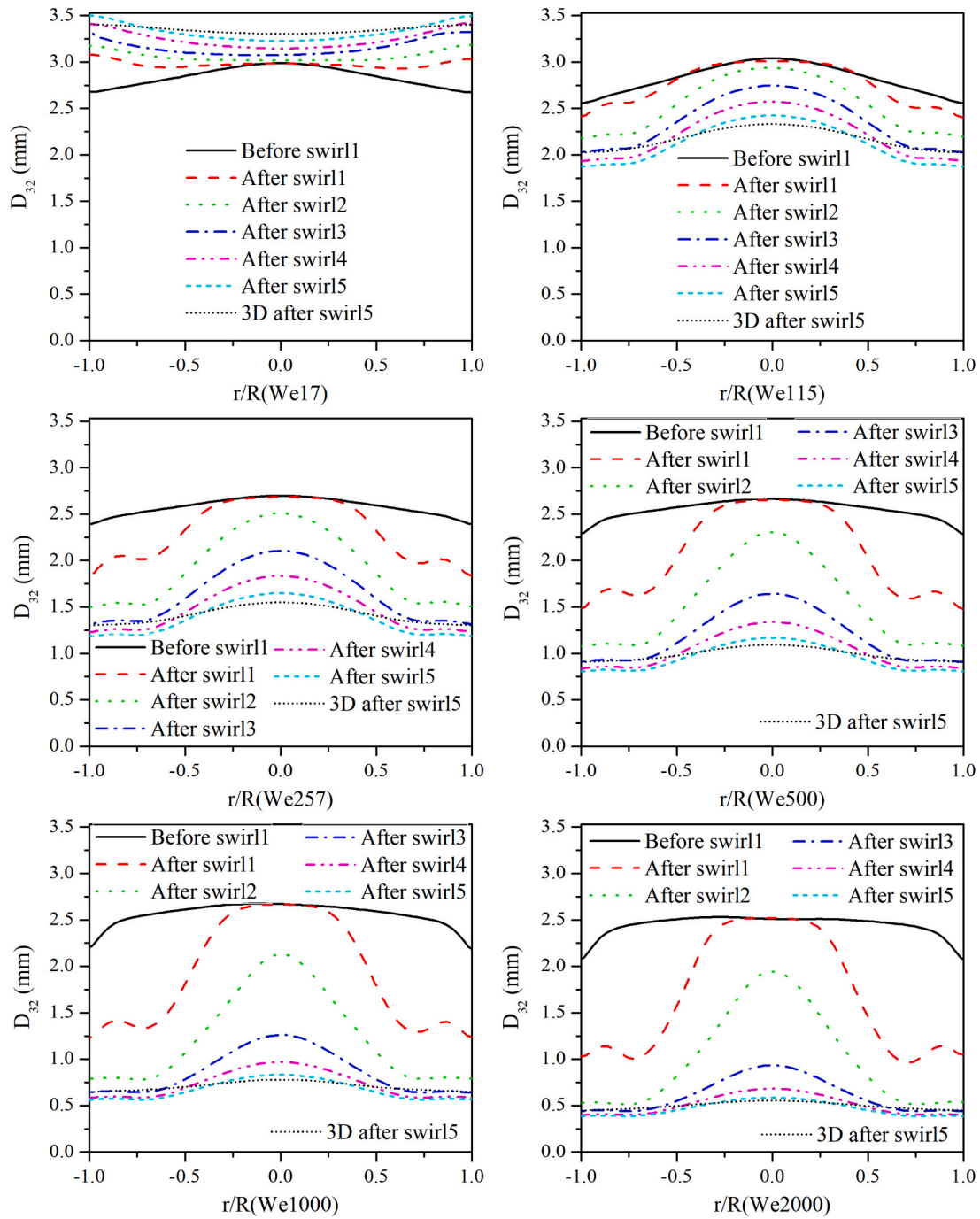


Fig. 7. Radial variation of the Sauter mean diameters at the planes to the entrance of the first swirl section, at each swirl section exit, and at 3D downstream of the last swirl section.

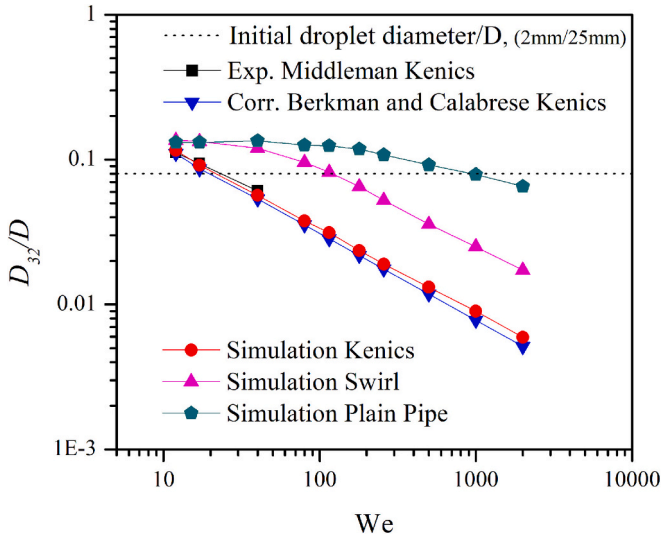


Fig. 8. Comparison of predicted Sauter mean diameter in the Kenics mixer, static swirl mixer and plain pipe.

correlate the DSD in a Kenics mixer. We will use the same equation to seek a correlation in the form of $F_v(D/D_{32})$ for the static swirl mixer. The cumulative distribution data in Fig. 10 are normalized to their mean D_{32} for each individual set and are replotted in Fig. 11. A fit of the data with MATLAB gives the cumulative distribution function as

$$F_v(X) = 0.5 \left[1 + \operatorname{erf} \left(\frac{X - 0.91}{0.38\sqrt{2}} \right) \right] \quad (24)$$

The root mean square deviation of the relative error between the data and the predicted values by CDF is 25 %, which is similar to Berkman and Calabrese's [14] correlation for the Kenics mixer. The goodness of fit of Eq. (23) is visualized in Fig. 11. In Middleman's [13] correlation for the Kenics static mixer, the equilibrium data were correlated by Eq. (23) with $\bar{X} = 1.06$ and $\sigma_v = 0.25$. Berkman and Calabrese [14] reported $\bar{X} = 1.12$ and $\sigma_v = 0.31$ for their Kenics mixer. The data for the static swirl mixer shows a slightly smaller mean and a slightly larger standard deviation. Despite the differences (e.g., in dispersion mechanisms, pipe dimensions, material properties, Weber number ranges, etc.), this correlation is very close to those reported for the Kenics mixers in the literature. This approach is, therefore, considered acceptable for evaluating the equilibrium DSD produced by the static swirl mixer.

3.2.3. Effect of number of swirls

Fig. 12 shows the mean droplet diameter along the static swirl mixer consisting of 15 swirl sections. The mean droplet diameters at the exit of each swirl section are calculated for the Weber numbers 257, 500, and 1000 respectively and are plotted in terms of the number of swirls. $n_s = 0$ is the condition before the swirl section, and $D_{32}(15)$ is the mean droplet diameter at the exit of the 15th swirl section. As shown in Fig. 12, the Sauter mean droplet diameter, D_{32} , decreases quickly to a very low level in the first 5 swirl sections as depicted by the steep slope, accounting for 86 %, 93 % and 96 % of the droplet size reduction for Weber numbers 257, 500 and 1000, respectively. Further size reduction in the 5th to 10th swirl sections is less prominent with the droplet size being reduced further by 11 %, 6 % and 3 % for Weber numbers 257, 500 and 1000, respectively. In the 10th to 15th swirl sections there is negligible change in size reduction, considering the extra pressure loss incurred. It would appear that 10 swirl sections are sufficient to produce the equilibrium drop size distribution, and 5 swirl sections are more cost-effective for the Weber number above 257.

3.2.4. Effect of pitch to diameter ratio

The aforementioned baseline static swirl mixer consists of 5 alternate clockwise and anticlockwise twisted swirl sections with a Pitch to Diameter (PD) ratio of 3, resulting in the same twist angle as the helical blades used in the Kenics mixer. This section attempts to examine the effects of different PD ratios on the droplet size reduction and the DSD of the static swirl mixer. As shown in Fig. 13, additional PD ratios of 5, 1.5 and 1 are simulated with identical conditions as the baseline static swirl mixer with a PD ratio of 3. The length of the mixers is kept the same as the Kenics mixer so as to compare the pressure loss incurred, resulting in 3, 5, 10 and 15 swirl sections for PD ratios of 5, 3, 1.5 and 1, respectively. The droplet size variation at a Weber number of 500 for various PD ratios shows that a smaller PD ratio promotes the formation of smaller droplets with the Sauter mean diameters at the plane, 3D downstream of the static swirl mixers, being 1.35 mm, 0.89 mm, 0.66 mm and 0.56 mm for PD ratios of 5, 3, 1.5 and 1, respectively.

The cumulative DSD of the droplets produced by the different PD ratios at a Weber number of 500 is depicted in Fig. 14, demonstrating that a reduced PD ratio leads to a smaller D_{32} and a narrower DSD, shifted towards smaller sizes. However, the difference between PD1.5 and PD1 is small in terms of mean size and DSD, suggesting that a PD ratio smaller than 1.5 is not notably beneficial. It can also be observed from Fig. 13 that, despite the different PD ratios, 10 swirl sections would appear to be sufficient to produce the equilibrium drop size distribution.

Fig. 15 shows the comparison of the final Sauter mean droplet diameter at planes 3D downstream of the static swirl mixer with various PD ratios, the Kenics mixer, and the plain pipe for a range of Weber numbers. The Sauter mean droplet diameter decreases with decreasing PD ratio for the static swirl mixer indicating enhanced dispersive mixing as stronger disruptive forces are induced by swirling flows from smaller PD ratios. It also depicts that the start points where the droplet breakage begins to dominate shifts to a smaller Weber number with reducing PD ratios, where the Weber numbers required to initiate the droplet size reduction are roughly at 200, 115, 65 and 50 respectively for PD ratios of 5, 3, 1.5 and 1, respectively. This is expected as the smaller the PD ratios, the stronger the turbulent dissipation rate. Despite the improved droplet size reduction effects with decreasing PD ratios, the static swirl mixers are still less effective than the Kenics mixer, roughly accounting for, on average, 50 %, 68 %, 78 % and 85 % of the Kenics mixer's size reduction for the same Weber numbers within their respective effective size reduction ranges.

3.2.5. Pressure loss

The pressure loss is an important parameter measuring the energy required for the operation of a static mixer in a pipeline. Typically, the formation of smaller droplets is at the expense of higher pressure losses as more energy is required to break up the droplets. Fig. 16 compares the friction factors of the static swirl mixers with various PD ratios, the Kenics mixer, and the plain pipe for a range of Weber numbers. The friction factor is higher for static swirl mixers with small PD ratios as higher pressure losses occur due to the increased wetted surface area and the improved dissipation of dynamic pressure of fluid near the tube wall. However, that aside, the static swirl mixers are essentially "open pipes" without internal insets and the friction factor is 75 %, 65 %, 46 % and 38 % lower than the Kenics mixer (which has internal structures) for the same Weber numbers for PD ratio of 5, 3, 1.5 and 1, respectively.

Therefore, adopting different PD ratios, the static swirl mixer, which has no inserted element inside and thus less costly to manufacture and maintain, is predicted to archive 50 % to 85 % of the droplet size reduction effects of the Kenics mixer, while expending 75 % to 38 % less pressure loss for the same Weber numbers. To achieve the same level of droplet size reduction as the Kenics mixer, the static swirl mixer needs to operate at a relatively higher Weber number. The desired Weber numbers for each PD ratio to achieve the same mean droplet size as the Kenics mixer can be calculated from the correlations between D_{32} and the Weber number. The corresponding pressure losses at the desired

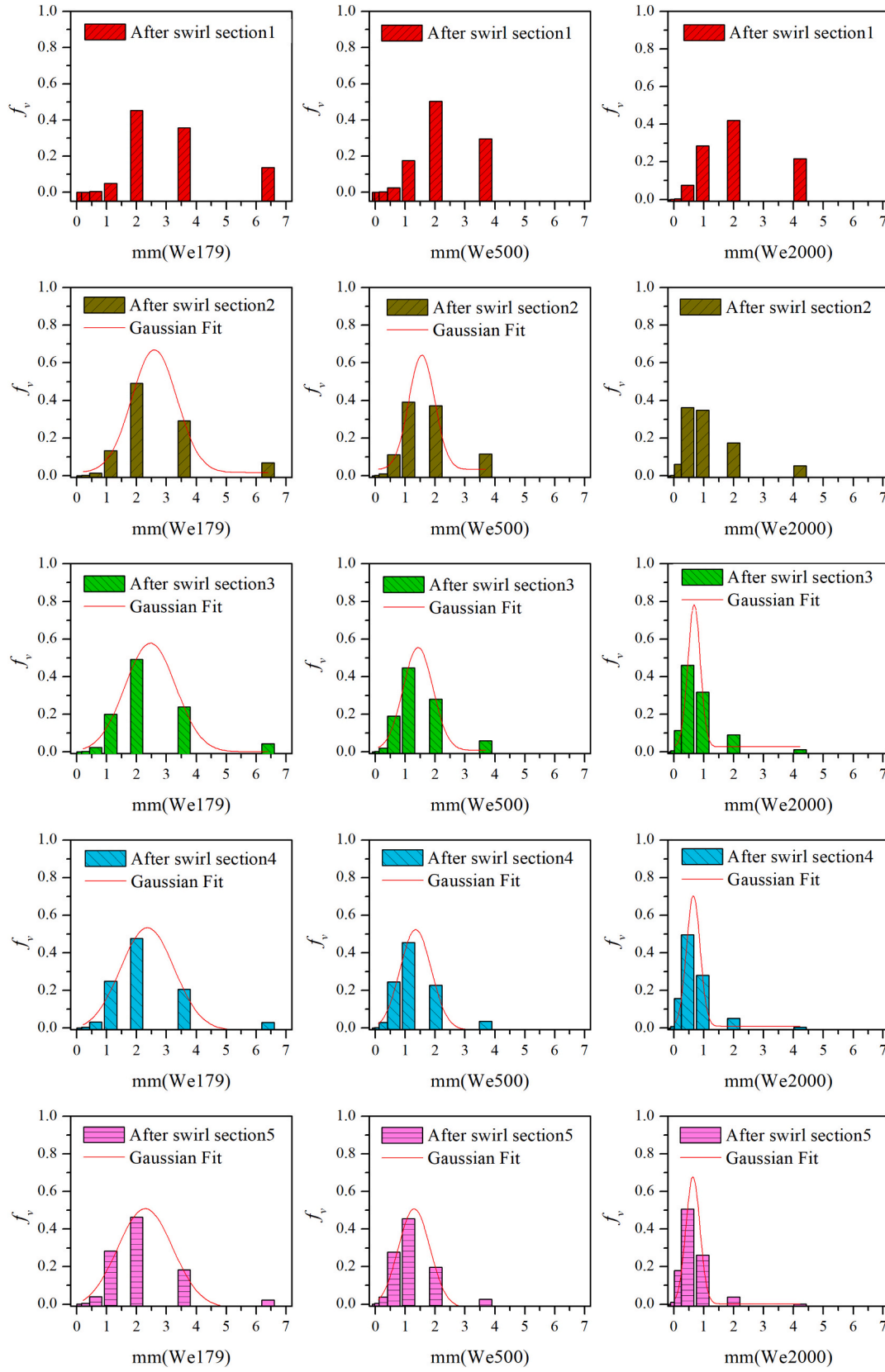


Fig. 9. Volumetric DSD at the planes of the exit of each swirl section for various Weber numbers.

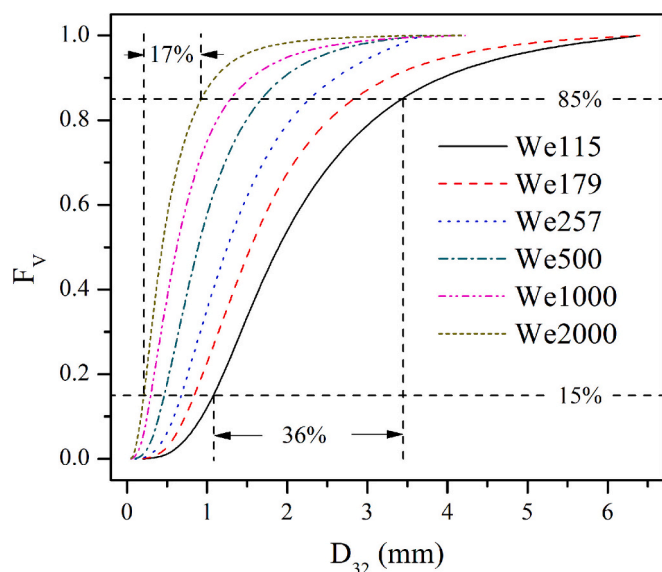


Fig. 10. Cumulative distribution of the final droplet size at the plane 3D downstream of the static swirl mixer for various Weber numbers.

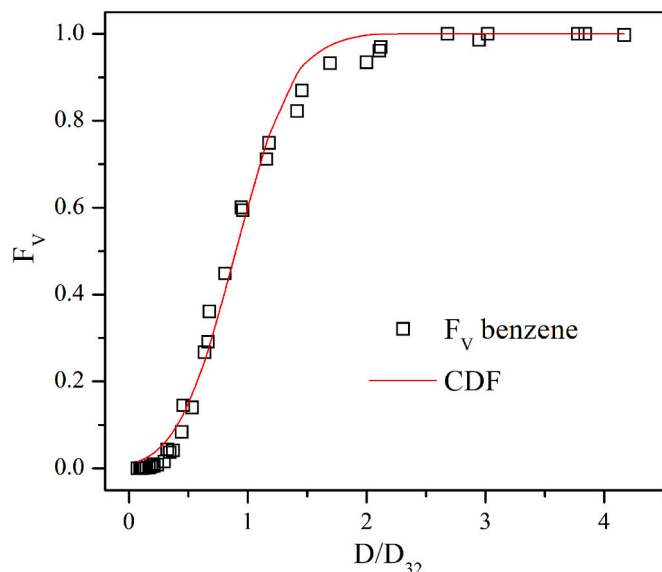


Fig. 11. Goodness of fit with Wang and Berkman's [42] cumulative distribution function (derived from immiscible liquid dispersion in stirred tanks, Weber number ranges 44 to 1137).

Weber numbers are estimated from the simulated pressure losses by interpolation. Fig. 17 gives a comparison of the pressure losses incurred solely by the Kenics mixer and static swirl mixers for various PD ratios, in producing the same D_{32} . Lower pressure losses of 32 %, 42 %, 22 % and 15 % were recorded, respectively, for static swirl mixers with PD ratios of 5, 3, 1.5 and 1 in comparison to that of the Kenics mixer, suggesting advantages in saving pumping energy. It is interesting to observe that the best energy saving takes place at around the PD ratio 3. A PD ratio larger or smaller than this will become less energy effective. The relatively larger pressure loss for a PD ratio of 5 may be attributed to its weaker size reduction effect, which needs to operate at much larger Weber numbers, causing a greater pressure loss than the PD ratio of 3. With decreasing PD ratio, the size reduction effect improves, but is also at the expense of higher pressure loss due to the increased wetted surface area and the improved dissipation of dynamic pressure of fluid near the tube wall, diminishing its cost-effectiveness. At the PD ratio of 1.5, the

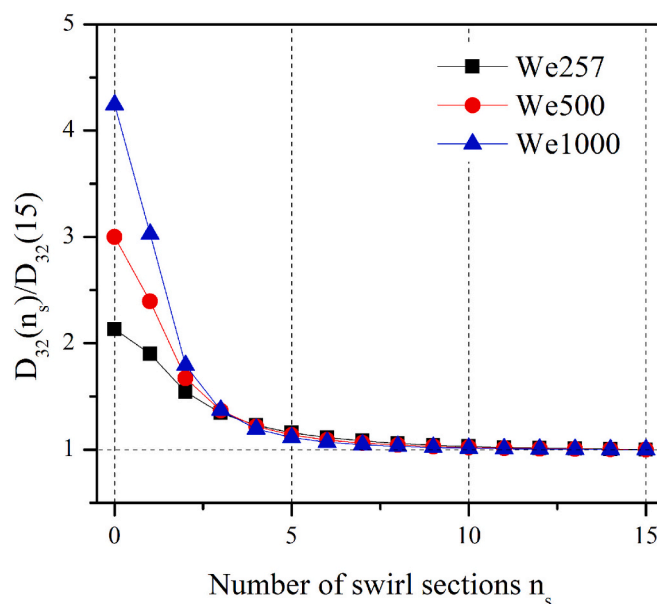


Fig. 12. Effect of number of swirl sections on droplet Sauter mean diameter.

static swirl mixer is still more energy efficient than the Kenics mixer at all the Weber numbers examined, with a mean estimated energy saving of 22 %. Further decreasing the PD ratio from 1.5 to 1 will lose its advantages at larger Weber numbers (indicated by the smaller D_{32}/D), where the pressure loss becomes very close to that of the Kenics mixer. Therefore, a PD ratio smaller than 1.5 is probably unnecessary considering the greater pressure loss incurred. At a ratio larger than 3, the size reduction effect is further weakened, and the overall length of the static swirl mixer will be longer if the same number of swirl sections are to be used, lowering its energy efficiency and applicability (e.g., require a larger space). It would appear that the optimal PD ratio lies between 1.5 and 3, which offers a balance between size reduction effect and pressure loss. Besides the lower pressure loss for mixing, the static swirl mixer has no mixing element mounted inside the pipe, such as helical blades or corrugated plates, that would contribute to problems of fouling and cleaning. In addition, the static swirl mixer has been proven to have a self-cleaning effect which will reduce the potential of fouling and thus the cost of maintenance [24,25]. The proposed static swirl mixer, therefore, has the potential to serve as an alternative for dispersive mixing to address disadvantages associated with mixing inserts, such as higher pressure drops, greater potential for fouling and the relative difficulty of cleaning.

4. Conclusions

In the present study, CFD-PBM model has been employed to investigate the performance of a static mixer without inserted elements for dispersion of immiscible liquids. The model was firstly validated in a Kenics mixer using available experimental data and empirical correlations and showed a good prediction accuracy. The CFD-PBM simulation on the proposed static swirl mixer revealed that it promotes a clear dispersive mixing, producing fine droplets at the expense of a lower pressure loss than the Kenics mixer under the same Weber number, attributed to the absence of inserted elements, but also less effective in size reduction than the Kenics mixer. In producing the same droplet sizes, the static swirl mixers cause a lower pressure loss than the Kenics mixer to a certain extent depending on different Weber numbers and pitch to diameter ratios, demonstrating an energy-saving potential. Stronger size reduction and narrower DSD are observed in flows with high Weber numbers. The detailed correlation of the final droplet Sauter mean diameters with Weber numbers, and the cumulative distribution

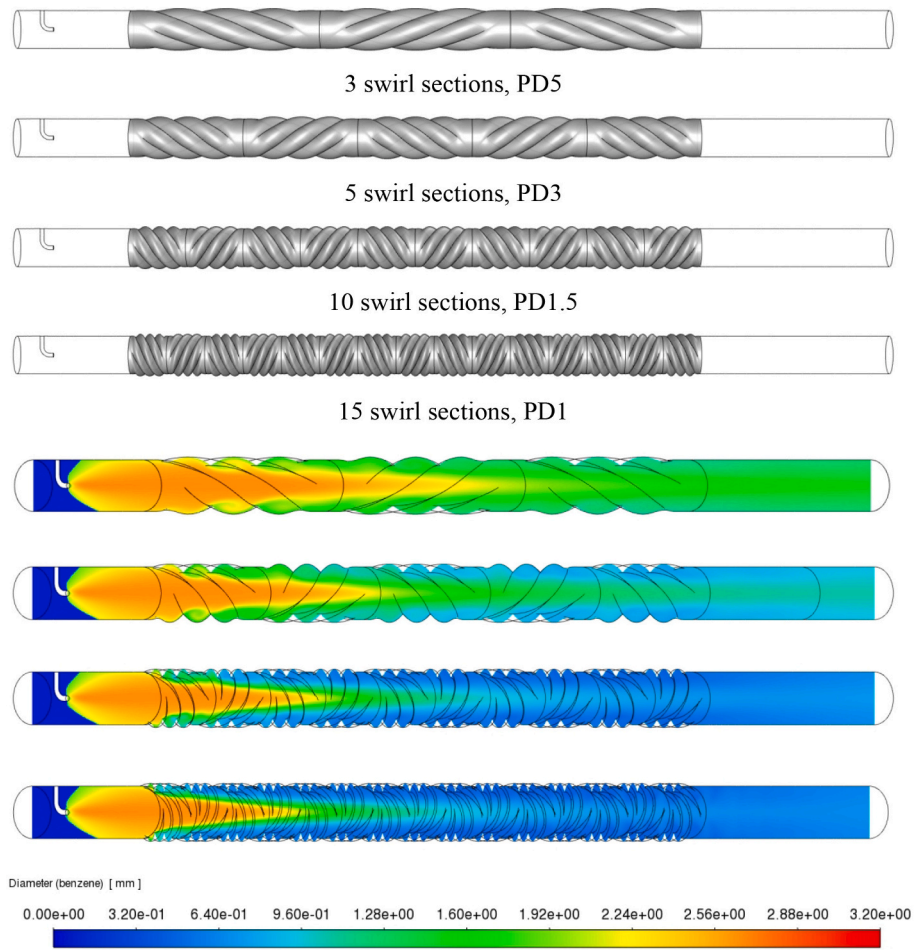


Fig. 13. Static swirl mixers with various PD ratio and their contour plot of D_{32} variation at We 500.

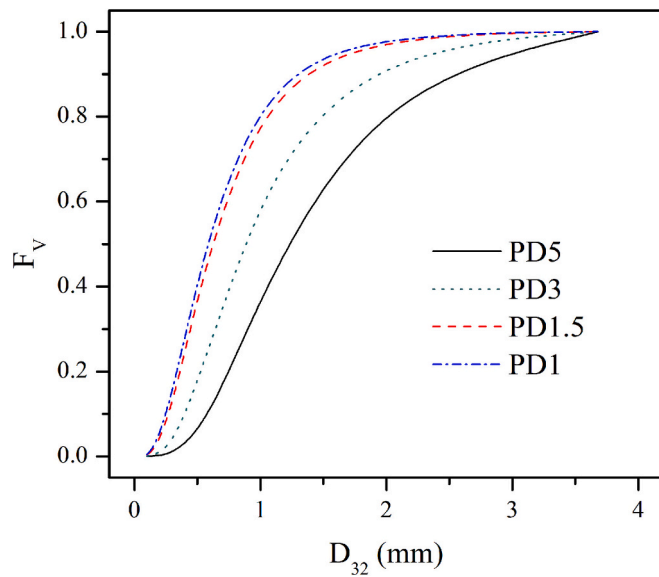


Fig. 14. Cumulative distribution of the final droplet size at the plane 3D downstream of the static swirl mixers with various PD ratios at a Weber number of 500.

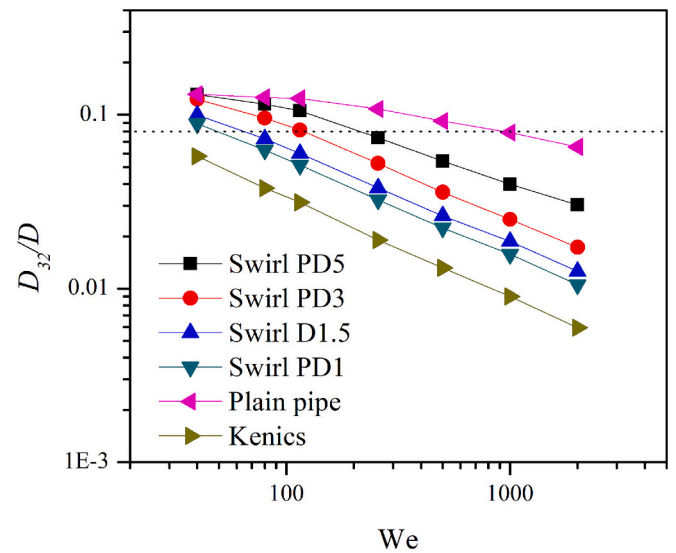


Fig. 15. Sauter mean droplet size at planes 3D downstream of the static swirl mixer with various PD ratios, the Kenics mixer, and the plain pipe.

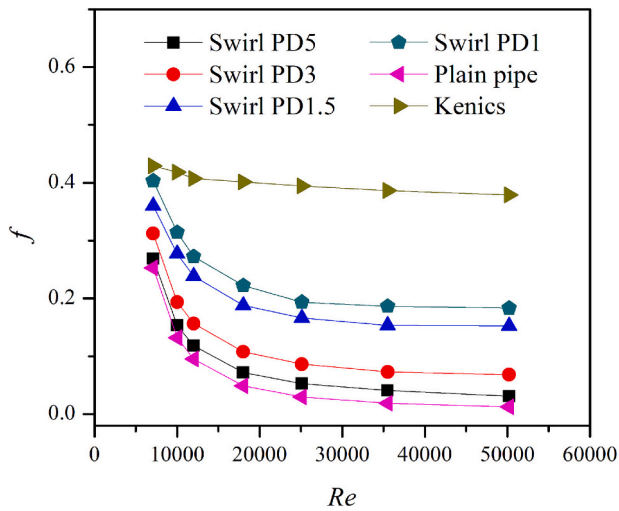


Fig. 16. Comparison of the friction factor in static swirl mixers for various PD ratios, the Kenics mixer, and the plain pipe for the same Weber numbers.

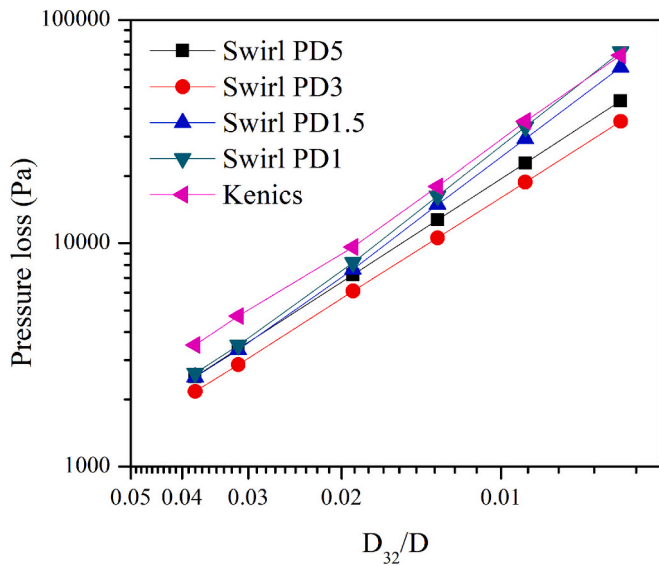


Fig. 17. Comparison of the pressure losses incurred by the Kenics mixer and the static swirl mixers for various PD ratios in producing the same D_{32} .

function of the droplet sizes under the present operation conditions are derived which provides insights into potential applications. The effects of PD ratio and the number of swirl sections on dispersion are investigated. A PD ratio between 1.5 and 3 seems to be optimal. Ten swirl sections would appear to be sufficient to produce the equilibrium droplet size distribution despite the different PD ratios. In light of the above findings, this study numerically demonstrated the potential of the proposed static swirl mixer, without inserted elements, to produce fine droplets with narrow distributions at the expense of lower energy consumption than mixers with inserted mixing elements mounted inside, e.g., the Kenics mixer. As the swirl sections are essentially an “open pipe” and has a self-cleaning effect [24,25], it has the potential to serve as an alternative to mitigate the disadvantages associated with the mixing inserts, such as high pressure drops, greater potential for fouling, relative difficulty of cleaning, and greater costs in materials and manufacturing.

Nomenclature

A	dimensionless empirical constant
a_{ij}	Reynolds-Stress anisotropy term
α_q	volume fraction of phase q
D	tube diameter, mm
D_{32}	Sauter mean diameter, mm
D_i	the i th droplet diameter, mm
D_k	Cumulative droplet diameter, mm
f	Fanning friction factor
f_v	volume fraction of a droplet size
F_v	Cumulative Distribution Function
G_V	growth rate based on particle volume
$g(V')$	breakage frequency
k	turbulence kinetic energy
L	length of pipe corresponding to the pressure loss, m
L_s	length of one mixing element, m
N	number of samples
n_e	number of mixing elements
n_i	the i th droplet
P	pressure, Pa
p	number of child particles produced per parent
R	tube radius, m
r	radial coordinate, m
S_q	source term
Re	Reynolds number
t	time, s
$\bar{U}_c$	continuous phase mean velocity, m/s
$\vec{v}_q$	velocity of phase q , m/s
Vi	viscosity group
V	particle volume
We	Weber number
$\vec{x}$	external coordinates of a particle's state vector
β	probability density function
ε	turbulence eddy dissipation rate, $m^2 \cdot s^{-3}$
μ_c	continuous phase viscosity, $Kg/(m \cdot s)$
μ_d	dispersed phase viscosity, $Kg/(m \cdot s)$
ρ_c	continuous phase density, $kg \cdot m^{-3}$
ρ_d	dispersed phase density, $kg \cdot m^{-3}$
σ	surface tension, $N \cdot m^{-1}$
σ_v	standard deviation
ϕ	internal coordinates of a particle
CFD	Computational Fluid Dynamics
DSD	droplet size distribution
PD	Pitch to Diameter ratio
PBM	Population Balance Model
RSM	Reynolds Stress Model

CRediT authorship contribution statement

Guozhen Li: Writing – original draft, Software, Methodology, Investigation. **Nicholas J. Miles:** Writing – review & editing, Supervision, Conceptualization. **Philip Hall:** Resources, Project administration, Funding acquisition. **Zheng Wang:** Investigation, Formal analysis. **Jiarui Gao:** Software. **Yubang Shen:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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